

Document id	Title	Organisation /Author	Date	Status
ITM-120507	mvdXML specification 1.0	Tim Chipman, Thomas Liebich, Matthias Weise	2012-05-11	Final

mvdXML

Specification of a standardized format to define and exchange
Model View Definitions with Exchange Requirements and Validation Rules

developed by
Model Support Group (MSG) of buildingSMART International Ltd.

authors
Chipman, Tim; Liebich, Thomas; Weise, Matthias

Version 1.0
14. May 2012

Document history

1.0	Final release of mvdXML. Accepted by bSI ITM committee as the official buildingSMART specification for publishing Model View Definitions	2012-05-14
0.9.4	The following changes were made in this draft: 1. EntityRule.EntityRules added for indicating subtype rules.	2012-05-11
0.9.3	The following changes were made in this draft: 1. ModelView.BaseView added for indicating add-on views. 2. ExchangeRequirement.applicability attribute added. 3. ConceptTemplate.ApplicableEntities renamed to ApplicableEntity.	2012-05-07
0.9.2	The following changes were made in this draft: 1. The ConceptLeafNode was renamed to Concept, with SubConcepts added. 2. ApplicableSchema attributes now use string instead of enumeration for version flexibility. 3. Cardinality includes “_asSchema” to indicate default cardinality.	2012-04-20
0.9.1	Combined mvdXML schema proposal incorporating the original mvdXML 0.5 with the proposed phase 2 extension, several simplifications: 1. The ConceptNode entity was deleted 2. The Concept abstract entity was deleted, since ConceptNode was deleted, where the only attribute was moved to the subtype ConceptLeafNode. 3. The ConceptRootNode Category attribute was deleted	2012-03-27
	Proposal for phase 2 of a formal mvdXML format: <i>Mapping of MVD concepts to IFC definitions</i> as appendix to mvdXML 0.5	2011-05-20
0.5	First buildingSMART release, no other changes of content	2011-06-19
0.4	Public release, first release after acceptance of mvdXML by buildingSMART ITM group, following changes have been made: 1. incorporation of the formally defined (IFC) schema, that describes the formal subschema corresponding to the Model View Definition 2. minor changes as result of first prototype developments	2011-05-05
0.3	Public release, incorporating feedback from buildingSMART MSG 1. Restructuring of document content, adding MVD history 2. adding general objectives, motivation and relation to MVD methodology 3. minor corrections in XSD Version 0.3 (key/keyref and href for Definition)	2011-03-04
0.2	Restricted release to buildingSMART MSG and TechCom, XSD Version 0.2	2011-02-16
0.1	Internal release, XSD version 0.1	2011-02-07

Table of Content

1	Overview.....	5
1.1	Purpose.....	5
1.2	Usage.....	5
2	Schema	6
3	mvdXML schema elements and types.....	7
3.1	Entity.....	7
3.2	Entity related definitions	7
3.2.1	Definition	7
3.2.2	Body	8
3.2.3	Link	8
3.3	mvdXML.....	8
3.4	ModelView	9
3.5	ExchangeRequirement.....	9
3.6	ConceptTemplate	10
3.6.1	AbstractRule	10
3.6.2	AttributeRule	10
3.6.3	EntityRule	11
3.6.4	TemplateRule	11
3.6.5	Constraint	12
3.7	ConceptRoot.....	13
3.8	Concept.....	13
3.8.1	Requirement.....	14
4	Examples.....	15
4.1	ModelView with Concepts but no Constraints	15
4.2	Concept Template with Constraints	20
4.3	Concept Template with Parameters.....	22
4.4	Concept Template with Referenced Values	23
4.5	Concept Root using Template Values.....	25
5	XSD Listing	26
6	Acknowledgment.....	34

Page no.	Authors
3	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

Table of Figures

Figure 1: Graphical representation of the mvdXML schema	6
Figure 2: Example diagram created for a concept template for 'Axis representation'	21
Figure 3: Example diagram created for a concept template for 'Axis representation'	22

Table of Tables

Table 1: Common attributes defined in the attributeGroup <i>identity</i>	7
Table 2: Common element references defined in the element <i>Definition</i>	7
Table 3: Common attributes defined in the element <i>Body</i>	8
Table 4: Common attributes defined in the element <i>Link</i>	8
Table 5: Common element references defined in the element <i>mvdXML</i>	8
Table 6: Common attributes and element references defined in the element <i>ModelView</i>	9
Table 7: Common attributes defined in the element <i>ExchangeRequirement</i>	9
Table 8: Common attributes and element references defined in the element <i>ConceptTemplate</i>	10
Table 9: Common attributes defined in the abstract element <i>AbstractRule</i>	10
Table 10: Common attributes and element references defined in the element <i>AttributeRule</i>	11
Table 11: Common attributes and element references defined in the element <i>EntityRule</i>	11
Table 12: Common attributes defined in the element <i>TemplateRule</i>	12
Table 13: Common attributes defined in the element <i>Constraint</i>	12
Table 14: Metric values and descriptions used in <i>Constraint</i>	12
Table 15: Operators used in <i>Constraint</i>	12
Table 16: Operators applied to types in <i>Constraint</i>	13
Table 17: Common attributes and element references defined in the element <i>ConceptRoot</i>	13
Table 18: Common element references defined in the element <i>Concept</i>	14
Table 19: Common element references defined in the element <i>Requirement</i>	14

1 Overview

A Model View Definition (MVD) describes contents of data to be exchanged in specific scenarios. mvdXML refers to an electronic format for representing such model view definitions. The purpose of this document is to describe the structure and usage of mvdXML.

While a data schema describes available data structures and type information, a model view definition describes graphs of such data structures to be used in particular scenarios with particular constraints. While mvdXML is a generic structure that could be applied to any data schema, the primary intention and documentation herein describes its use relative to the Industry Foundation Classes (IFC) data schema, the ISO16739 standard.

1.1 Purpose

The mvdXML format serves several purposes:

- To support automated validation of IFC data sets for quality assurance and certification.
- To generate documentation for specific model views and the IFC specification itself.
- To support software vendors providing filtering of IFC data based on model views.
- To limit the scope of IFC to well-defined subsets applicable for particular applications.

1.2 Usage

While it is still being possible to write an mvdXML based Model View Definition by using any text editor, it is anticipated that specific software applications are used to read and write mvdXML data sets. Software for working with mvdXML may include the following.

- The IFC Documentation Generator (IFCDOC.EXE), is a free tool issued by buildingSMART, that reads and writes mvdXML, and provides a graphical user interface for defining all content within mvdXML. It can be preloaded with a particular IFC release specification and allows access to all parts of the IFC specification when developing the mvdXML concepts and constraints. This tool may also auto-generate instantiation diagrams, output HTML documentation for model views, and is also used for generating the IFC4 documentation.
- XML/XSD editors such as Microsoft Visual Studio and Eclipse may edit mvdXML in raw format, just as any other XSD-based schema.
- Testing servers may read mvdXML and use such information to validate submitted IFC files for conformance. The IFC certification server, GTDS, that is used for the official certification of the IFC2x3 coordination view v2.0 by buildingSMART International has the capability to import mvdXML to set up the structural and rule sets for a new certification process.
- IFC-based software applications may read mvdXML for automatically filtering and validating data to conform to the specified constraints. It is also possible for IFC-based software applications to write mvdXML to enable users to define custom exchange scenarios.

Page no.	Authors
5	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

2 Schema

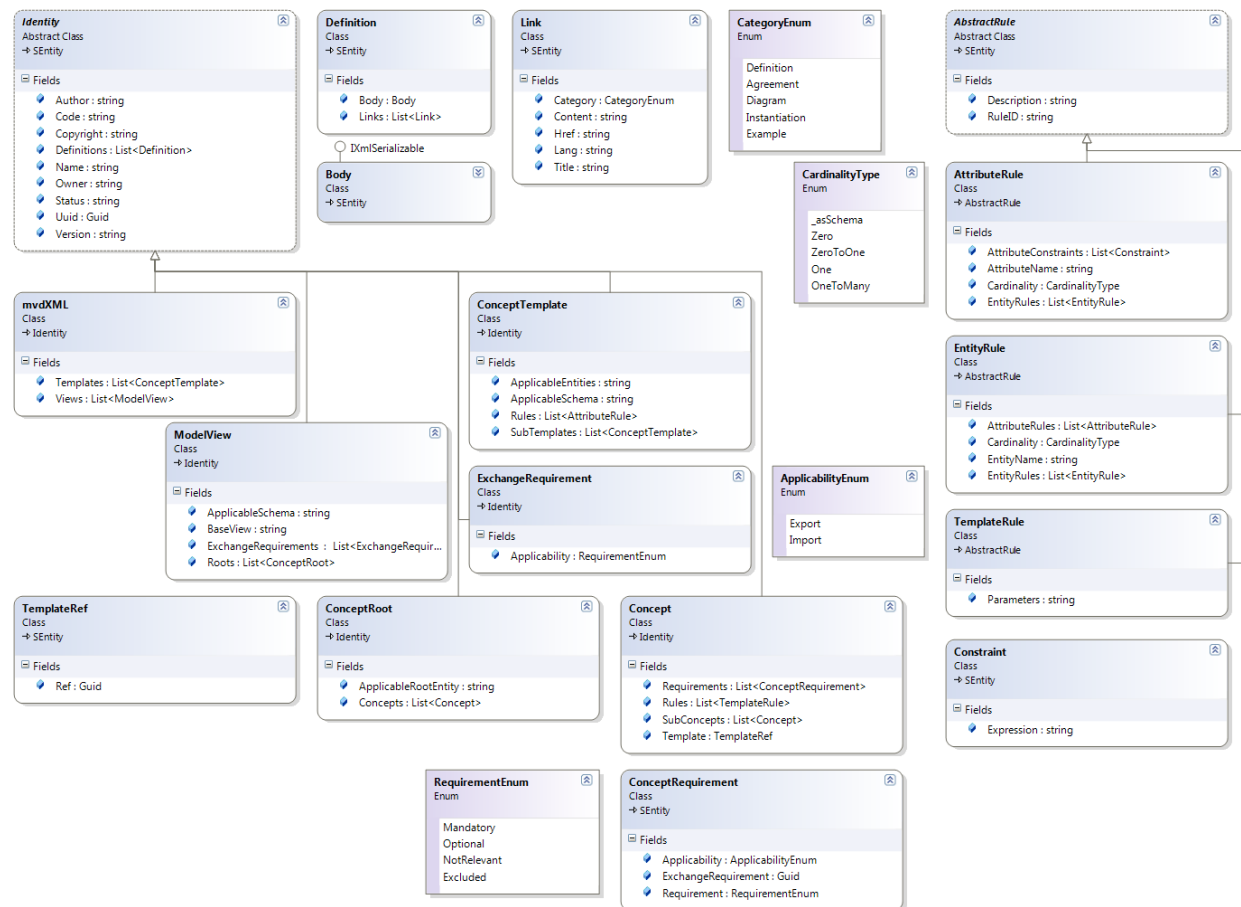


Figure 1: Graphical representation of the mvdXML schema

The basic structure of a model view definition includes a set of root concepts, *mvd:ConceptRoot*, each of which references a specific IFC entity, e.g. *IfcWall*, representing a major and individually testable element in a MVD. Each root concept contains a set of concepts, *mvd:Concept*, which describe rules for common subsets of information (e.g. material usage) within the context of the particular root. Each concept is backed by a template, *mvd:ConceptTemplate*, describing a graph of object instances, relationships, and constraints, where the concept may provide a set of parameters that apply to the referenced template. Each concept template may describe a graph of instances indicating required values for attributes and graphs of entity references. Such values may be constrained with comparison operators to indicate literal values or reference values.

A model view indirectly describes the set of IFC entities within scope based on entities referenced from each root concept, and entities or types (and subtypes as defined) used within instance graphs. All such IFC entities may be combined and published as the IFC schema subset corresponding to the model view. The IFC schema subset has to be a valid schema by itself.

A model view also includes a set of exchanges describing particular scenarios where information is exchanged from one role to another; for example an architect providing design information to a structural engineer. Each concept may have requirements set for each exchange, indicating mandatory, optional, or restricted usage.

Page no.	Authors
6	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

3 mvdXML schema elements and types

This section introduces the individual mvdXML entity, type and attribute definitions.

3.1 Entity

Similar to IFC, the mvdXML schema makes a distinction between elements having identity and those that do not. All elements with identity have the following attributes and sub elements defined.

NOTE The mvdXML.xsd does not incorporate an *mvd:Identity* abstract class, the common attributes are defined in the attributeGroup name="identity" and the definitions in the element name="Definition".

Attribute	Type	Description
uuid	Guid	Universally unique identifier. This is used as a persistent identifier, and must never change.
name	String	Human readable name. This is used as the header of the section and entry within table of contents when generating documentation.
code	String	Human readable reference value.
version	String	Sequential version number of the concept.
status	String	The status of the concept (Sample, Draft, Proposal, Candidate, Official or Deprecated).
author	String	The author(s) of the work.
owner	String	The owner of the work.
copyright	String	The copyright under which the work is published. Note: if adopted by buildingSMART international, the copyright shall lie either with buildingSMART International, or is governed by a well-recognized open license (e.g. creative commons, open source BSD/ GNU).

Table 1: Common attributes defined in the attributeGroup *identity*.

3.2 Entity related definitions

The *mvd:Definition*, and the elements referenced by the element *Definition*, *mvd:Body*, and *mvd:Link* elements provide the capability to add multi-lingual descriptions at any element with own identity. Such elements are:

- *mvd:mvdXML*
- *mvd:ModelView*
- *mvd:ExchangeRequirement*
- *mvd:ConceptTemplate*
- *mvd:ConceptRoot*
- *mvd:Concept*

3.2.1 Definition

The element *mvd:Definition* groups definition text and links to additional figures, diagrams, examples, and other external documents.

Attribute	Type	Description
Body	Body [0:1]	HTML-formatted description of the concept in the default language.
Links	Link [0:?]	List of additional content, each of which may be in separate languages.

Table 2: Common element references defined in the element *Definition*.

3.2.2 Body

The element *mvd:Body* holds the definition text that is qualified by a language tag.

NOTE In order to correctly encapsulate the HTML formatted text, the content shall be tagged by `<![CDATA[ ]>` to preserve the HTML code.

Attribute	Type	Description
lang	String	Locale identifier (e.g. 'en', 'de', 'en-GB', 'de-CH') or blank to indicate default locale.
(content)	String	HTML-formatted content for generating documentation. Content within should be encapsulated by paragraph tags (" <code><p></code> ") and/or list tags (" <code></code> "). Images should not be contained within; rather they should be specifically referenced by the Link element (allowing for automatic figure numbering).

Table 3: Common attributes defined in the element *Body*.

3.2.3 Link

The element *mvd:Link* holds all links to additional documentation content.

Attribute	Type	Description
lang	String	Locale identifier (e.g. 'en', 'de', 'en-GB', 'de-CH') or blank to indicate default locale.
category	String	definition: formatted as documented definition in alternate locale; agreement: formatted as NOTE in documentation; diagram: formatted as custom figure in documentation based on href; instantiation: formatted as instance diagram figure based on href; example: formatted as EXAMPLE in documentation based on href.
title	String	Human readable name. This is used as the header of the section and entry within table of contents when generating documentation.
href	String	Relative URL to referenced content, particularly for diagrams and examples that are manually generated. (Diagrams that are auto-generated from rules are not included.) This is used to reference any external files such that they are included when generating documentation.
(content)	String	HTML-formatted description in specified language.

Table 4: Common attributes defined in the element *Link*.

3.3 mvdXML

This element comprises the scope of the mvdXML document; it includes zero-to-many *mvd:ModelView*'s and one-to-many *mvd:ConceptTemplate*'s. It is recommended to include all concept templates that are referenced by included model view(s), or else distribute the mvdXML file along with other mvdXML files containing such templates.

NOTE A particular usage of mvdXML is to only publish concept templates. In this case, the *ModelView* element remains empty.

Attribute	Type	Description
Templates	ConceptTemplate [0:?]	Set of templates, which may be exchanged with or without referencing model view definitions.
Views	ModelView [0:?]	List of model views definitions, in order of listing in generated documentation.

Table 5: Common element references defined in the element *mvdXML*.

3.4 ModelView

This element is the description of a Model View Definition (MVD); it is specific to an IFC schema release and contains one-to-many *mvd:ConceptRoot*'s. It also includes the reference to zero-to-many applicable *mvd:ExchangeRequirement*'s. Multiple model views from potentially different releases may be contained in the same file.

The set of entities and types regarded to be within scope of a model view is not explicitly defined; rather it is indirectly determined by constructing a graph of *mvd:ConceptRoot* elements and following the set of rules indicating referenced entities within scope. Thus, describing the set of rules automatically determines what is in or out of scope, preventing the possible mismatch of missing data structures that are required, or included data structures that are not documented for use.

EXAMPLE The "CoordinationView_V2.0" is a Model View Definition, it is captured by a *mvd:ModelView* element. It has the @name="CoordinationView_V2.0", the @applicableSchema="IFC2X3", and a reference to the four exchange requirements currently defined for the Coordination View Version 2.0.

Attribute	Type	Description
BaseView	uuid anyURL	Reference to a base model view definition (in case that this model view represents an add-on model view that extents a base view.
applicableSchema	String	Identifies the schema using the ISO 10303 schema identifier, such as IFC2X_FINAL, IFC2X2_FINAL, IFC2X3, or IFC2X4.
ExchangeRequirements	ExchangeRequirement	List of exchanges within the model view, in logical order.
Roots	ConceptRoot	Set of root concepts within scope of the model view.

Table 6: Common attributes and element references defined in the element *ModelView*.

3.5 ExchangeRequirement

This element is the description of an Exchange Requirement Model (ERM) that is covered by the MVD. An ERM covers the Exchange Requirements (ER) that are identified for a particular exchange scenario that is covered by the MVD. ERM's may add additional constraints to the use of concepts and are an important part of later certification and validation processes.

An ERM can be referenced from a *mvd:Concept* to impose specific constraints for exchanges that reference this ERM. An ERM can be specifically declared to be only applicable for import or export scenarios using the attribute *applicability*.

EXAMPLE The ERM "Architecture" capturing the ER for exporting an architectural building model is an exchange requirement model within the CoordinationView_V2.0. It is captured by a *mvd:ExchangeRequirement* element. It has the @name="Architecture", and the @applicability="export".

Attribute	Type	Description
applicability	Enum: import, export	Identifies if the ERM is specific for import, export or both. If such value is provided, then any referencing requirements must match; for example, if such value indicates <i>export</i> , then referencing requirements may use <i>export</i> but not <i>import</i> ; if such value is not provided, then referencing requirements may use any value.

Table 7: Common attributes defined in the element *ExchangeRequirement*.

3.6 ConceptTemplate

This element represents the reusable concepts; it may have one-to-many sub *mvd:ConceptTemplate*'s and thereby may form a tree of related reusable concepts. Each *mvd:ConceptTemplate* has an applicable schema and may have applicable root entities (i.e. concept roots to which the *mvd:ConceptTemplate* applies).

NOTE For buildingSMART compliant MVD documentation generation, each *mvd:ConceptTemplate* appears in Chapter 4 of the resulting HTML based documentation, with descriptive text and a diagram generated from rules.

EXAMPLE Decomposition (the re-usable concept of decomposing elements into parts)

Attribute	Type	Description
applicableSchema	enum: IFC2x, IFC2x2, IFC2x3, IFC4	Identifies the default schema for which the template applies. The template may be used for model views of other schemas, if all enclosed rules resolve to available attributes and types. .
applicableEntity	String	Indicates the IfcRoot-based entities for which the concept applies. It is recommended to use a single base class (e.g. IfcElement). This value provides the context for any attribute rules and is used within MVD tools to filter the list of available templates for particular entities. For a subtemplate, the applicable entity must be the same type or a subtype of the outer template. This value may be blank to indicate an abstract template that cannot be instantiated, containing sub-templates for specific entities.
SubTemplates	ConceptTemplate [0:?]	Set of sub-templates, potentially having a subset of applicable entities, which further define a concept template for particular usage. For example, a template for material usage may have sub-templates for material layer sets, material profile sets, and material constituent sets.
Rules	AttributeRule [0:?]	Set of attributes defined at applicableEntities, where each attribute may have value constraints and/or graphs of object instances defined. If an attribute is not defined, then the requirements are the same as indicated for the schema. <i>[v1.0] Restricted to AttributeRule (previously AbstractRule) for uniform usage.</i>

Table 8: Common attributes and element references defined in the element *ConceptTemplate*.

3.6.1 AbstractRule

This abstract element represents parameters in common with all rules.

Attribute	Type	Description
RuleID	String	Identifies the rule for referencing at templates.
Description	String	Description of the rule.

Table 9: Common attributes defined in the abstract element *AbstractRule*.

3.6.2 AttributeRule

This element represents the specification of an attribute on an entity, with related constraints, and/or entity rules.

Attribute	Type	Description
AttributeName	String	The case-sensitive name of the attribute relative to the enclosing EntityRule (if exists) or the enclosing applicableEntities of the ConceptTemplate.
Cardinality	enum: Zero, ZeroToOne, One, OneToMany, _asSchema.	For collection-based attributes (LIST or SET), indicates Zero if it must be empty; ZeroToOne if the collection may contain zero or one member; One if the collection must contain one member; OneToMany if the collection may contain one or more members; or _asSchema to indicate the behavior is the same as the schema (typically 0-N).

EntityRules	EntityRule [0:?]	An empty list indicates that any type may be used according to the schema. If one or more entities or types are defined, then instances must match one of the entries. The list of entries is expanded by each referencing TemplateRule defined at a Concept, where downstream rules apply according to the matching entity rule.
Constraints	Constraint [0:?]	Set of expressions, which all must evaluate to True for the referenced attribute. This implies a Boolean AND combination, without support for other Boolean operators (e.g. OR, XOR), however a future mvdXML version may include such capability.

 Table 10: Common attributes and element references defined in the element *AttributeRule*.

3.6.3 EntityRule

This element represents the specification of an entity (or value type) referenced by an attribute, either as a scalar reference or a reference from within a collection.

Attribute	Type	Description
EntityName	String	The case-sensitive name of the entity (e.g. "IfcBeam") which must be assignable to the enclosing AttributeRule (i.e. entity subtype or select member).
Cardinality	enum: Zero, ZeroToOne, One, OneToMany, _asSchema.	Indicates Zero if it must not contain any elements of the specified type; ZeroToOne if the collection may contain zero or one member of the specified configuration; One if the collection must contain one member of the given configuration; OneToMany if the collection must contain one or more members; or _asSchema for no requirement.
AttributeRules	AttributeRule [0:?]	Indicates a list of attributes constrained on the referenced ENTITY. If an attribute is not defined, then the requirements are the same as indicated for the schema.
EntityRules	EntityRule [0:?]	Indicates a list of entities deriving from the specified entity (directly or indirectly) that may have more specific rules defined. If no list is provided, then this rule applies to the referenced entity (according to <i>EntityName</i>) and all subtypes. If such list is provided, then this rule applies only to the referenced entity (not subtypes), where such entries in the list indicate usage for referenced subtypes.
Constraints	Constraint [0:?]	Set of expressions, which all must evaluate to True for the referenced attribute. This implies a Boolean AND combination, without support for other Boolean operators (e.g. OR, XOR), however a future mvdXML version may include such capability.

 Table 11: Common attributes and element references defined in the element *EntityRule*.

3.6.4 TemplateRule

This element represents an instantiation of a rule with specified parameters. It allows repetitive definitions to be efficiently represented, such as lists of applicable ports, materials, units, property sets, etc. The *@RuleID* of the template rule serves as a reference to the *@RuleID* of an *mvd:EntityRule* at the referenced *mvd:ConceptTemplate* to be instantiated. For example, if rules are defined for applicable ports on a device, the *@RuleID* would refer to an *mvd:EntityRule* linked to *IfcDistributionPort*.

If the referenced *mvd:EntityRule* is part of a SET-based attribute, then the instance is required to uniquely exist once (having unique combination of defined parameters), but without regard for order (as a SET has no implied order). If the referenced *mvd:EntityRule* is part of a LIST-based attribute, then the instance is required to occur at the relative position of the *mvd:TemplateRule*.

Attribute	Type	Description
Parameters	String	Expression of zero, one, or more parameters in the form "{Parameter}={Value};" (similar syntax as HTML styles), where a semicolon delimits each key-value pair. The Parameter corresponds to the RuleID of an AttributeRule at the referenced ConceptTemplate. The Operator must be Equals (any other behavior must be defined on constraints for rules within the template). The Value uses the same syntax as defined at the Constraint element. If the Parameter refers to an AttributeRule that is inside the scope of the referenced EntityRule, then the attribute is used for validation; otherwise if the Parameter refers to an AttributeRule that is outside the scope of the referenced EntityRule, then the attribute is conditional. For example, on an IfcSensor port definition where the RuleID refers to IfcDistributionPort, the "PredefinedType" parameter refers to IfcSensor.PredefinedType having value of "SENSOR", and the "Name" parameter refers to IfcDistributionPort.Name having value of "Output", it indicates that a port must exist at the particular position with a Name equal to the specified value (as inner validation) IF the PredefinedType equals the specified value (as outer conditional).

 Table 12: Common attributes defined in the element *TemplateRule*.

3.6.5 Constraint

This element is defined within the elements *mvd:EntityRule* and *mvd:AttributeRule* and represents a restriction on an attribute, which may require the value, type, or collection size to have equality (or other comparison) to a literal value or referenced value.

Attribute	Type	Description
Expression	String	Each expression is in the form {Metric}{Operator}{Benchmark}. For example, "Value=1". The Metric and Operator may be omitted to indicate value equality; that is "1" and "Value=1" are equivalent. Future extensions may allow for combinations of Boolean operators, (AND, OR, XOR), etc., however this version only specifies a minimal format for initial interoperability.

 Table 13: Common attributes defined in the element *Constraint*.

Metric may be one of the following:

Metric	Description
Value	Indicates the value of the attribute (value uses syntax according to the attribute type, defined below).
Size	Indicates the size of a collection or STRING (value is an INTEGER).
Type	Indicates the type of the value assigned to the attribute (value is a STRING).
Unique	Indicates whether value must be unique within the population of instances described within the Concept Template (BOOLEAN).

 Table 14: Metric values and descriptions used in *Constraint*.

Operator may be one of the following:

Operator	XML Escaped	Description
=	=	Equal.
!=	!=	Not Equal.
>	>	Greater Than.
>=	>=	Greater Than Or Equal.
<	<	Less Than.
<=	<=	Less Than Or Equal

 Table 15: Operators used in *Constraint*.

Benchmark may be either a literal value or a referenced value. _The syntax of literal values varies according to the EXPRESS attribute type and follows ifcXML (ISO-10303-28) format:

Type	Operators	Description
INTEGER	=, !=, >, >=, <, <=	The integer value.
REAL	=, !=, >, >=, <, <=	The real value including decimal point, where equality is exact (no epsilon offset).
BOOLEAN	=, !=	The boolean value as "true" or "false".
LOGICAL	=, !=	The logical value as "true", "false", or "unknown".
ENUM	=, !=	The enumeration value by case-insensitive name.
STRING	=, !=, >, >=, <, <=	The string value, which may optionally be enclosed by single quotes (if escaping required). Comparison operators indicate alphabetical sorting (e.g. ">=" can indicate "must start with" such as the scope of a classification reference, or earliest date/time).
BINARY	=, !=	The binary value encoded as hexadecimal prefixed by "%" and number of unused bits.
ENTITY	=, !=, >, >=, <, <=	The name of the entity type (e.g. "IfcWall"). Equality means exact type match; ">" means subtype of; ">=" means same type or subtype; "<" means supertype of; "<=" means same type or supertype.

Table 16: Operators applied to types in *Constraint*.

Parameter values are indicated by specifying "#" followed by an arbitrary identifier of a shared instance within the same *mvd:ConceptTemplate*. For example, a rule could indicate for *IfcBeamStandardCase* that an *IfcProfileDef* for an *IfcExtrudedAreaSolid* must match the profile defined within an associated *IfcMaterialProfile*, where they each equal the same value referenced as "#MP". Refer to the example "Concept Template with Referenced Values" as an illustration of how to use such value references.

3.7 ConceptRoot

This element represents the root element (other terms are "leaf node class", "variable concept") that represent the fundamental parts of an MVD that is represented by a collection of supported concepts.

Attribute	Type	Description
applicableRootEntity	String	Identifies the IFC entity (deriving from IfcRoot) for which the concept applies. Note that non-rooted entities are described by referencing rules, as such instances cannot exist on their own where usage is always dependent upon the referencing IfcRoot-based instance.
Requirements	Requirement [0:?]	Set of requirements which describe applicability of the concept to particular exchanges for import or export.
Concepts	Concept [0:?]	List of concepts for the applicable root entity, where each concept is of a unique template. The order of elements indicates the sequence displayed in generated documentation.

Table 17: Common attributes and element references defined in the element *ConceptRoot*.

3.8 Concept

This element represents a use definition for a particular entity with specific rules to be enforced.

Attribute	Type	Description
Template	TemplateRef	Mandatory reference to the ConceptTemplate by GUID, where such template may be defined within the same file or an external file.
Requirements	Requirement [0:?]	Set of requirements which describe applicability of the concept to particular exchanges for import or export.

Rules	TemplateRule [0:?]	Set of rules indicating how template applies to particular entity. For example, a rule for property sets may list each applicable property set; a rule for ports may list the name, type, and direction of each port. For LIST types, multiple rules indicate a sequence of instances which must match the order of the rules. For SET types, multiple rules indicate a set of instances which ALL must be included according to the rules. For SELECT, ENTITY, and all other types, multiple rules indicate any valid state (it is valid for the attribute to reference objects that match one of the elaborated configurations). All rules are combined using AND logic implicitly. [v1.0] Restricted to TemplateRule (previously AbstractRule) to enforce re-usable concepts.
SubConcepts	Concept [0:?]	Optional list of sub-concepts.

Table 18: Common element references defined in the element *Concept*.

3.8.1 Requirement

This element represents a use definition for a particular entity with specific rules to be enforced.

Attribute	Type	Description
Applicability	enum: import, export	Whether requirement applies to import or export.
Requirement	enum: mandatory, optional, not relevant, excluded	mandatory: must be included; optional: may be included; not relevant: no requirement; excluded: must not be included.
exchangeRequirement	Guid	Identifies the ExchangeRequirement by GUID within the same Model View Definition.

Table 19: Common element references defined in the element *Requirement*.

4 Examples

4.1 ModelView with Concepts but no Constraints

The following basic example shows an excerpt from the model view definition of the IFC2x3 Coordination View Version 2.0. It includes a single *mvd:ModelView* with several *mvd:ExchangeRequirement*. Some *mvd:ConceptTemplate* are included that are referenced from *mvd:Concept* definitions.

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="../xslt/mvdXML V0-3 style-1.xslt"?>
<mvd:mvdXML name="buildingSMART international MVD's" uuid="4afb1a8b-0b61-4ff8-9863-c10690fe06f2" xmlns="http://buildingSMART-tech.org/mvdXML/mvdXML1-0"
xmlns:mvd="http://buildingSMART-tech.org/mvdXML/mvdXML1-0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://buildingSMART-tech.org/mvdXML/mvdXML1-0
../schema/mvdXML_V1-0.xsd">
  <mvd:Templates>
    <mvd:ConceptTemplate uuid="02eefcc6-b26c-4f20-9c7c-c30ae74eb9a9" code="bSI-100" name="Aggregation" version="1.0" status="draft"
      owner="buildingSMART International" applicableSchema="IFC2x3">
      <mvd:Definitions>
        <mvd:Definition>
          <mvd:Body lang="en">
            <![CDATA[<p>Provision of an aggregation structure (or decomposition tree) of physical elements, where the <em>&quot;element&quot;</em> is either the whole or one of the parts of
an aggregated structure.</p>
<p style="margin-left: 40px;">Note: The <em>Aggregation</em> concept is specific to the composition and decomposition of physical elements. There is no order applied among the
parts. Beside the decomposition tree structure, the sub concepts also determine the correct assignment of shape, geometry, property set and other information to either the whole
or the parts.</p>
<p>The shape representation of the whole is constructed from the sum of the shape representations of the parts.</p>]]>
          </mvd:Body>
          <mvd:Link lang="en" category="agreement" title="CV-2x3-119"
            href="http://buildingSMART-tech.org/developers/ifc-implementation/ifc-impl-agreements/cv-2x3-119" xsi:nil="true"/>
          <mvd:Link lang="en" category="agreement" title="CV-2x3-120"
            href="http://buildingSMART-tech.org/developers/ifc-implementation/ifc-impl-agreements/cv-2x3-120" xsi:nil="true"/>
          <mvd:Link lang="en" category="agreement" title="CV-2x3-121"
            href="http://buildingSMART-tech.org/developers/ifc-implementation/ifc-impl-agreements/cv-2x3-121" xsi:nil="true"/>
        </mvd:Definition>
      </mvd:Definitions>
    </mvd:ConceptTemplate>
    <mvd:SubTemplates>
      <mvd:ConceptTemplate uuid="f6c8ea4c-e0aa-422c-a852-a1711a20d228" code="bSI-101" name="Decomposition" status="draft" version="1.0"
        owner="buildingSMART International" applicableSchema="IFC2x3">
        <mvd:Definitions>
          <mvd:Definition>
            <mvd:Body lang="en">
              <![CDATA[<p>Provision of an aggregation structure where the <em>&quot;element&quot;</em>, representing the aggregate, is decomposed into parts represented by other elements. The
<em>&quot;element&quot;</em> establishes the common object coordinate system to which the parts are placed relative. The <em>&quot;element&quot;</em> shape representation is
constructed out of the shape representations of its parts.</p>
<p style="margin-left: 40px;">Note: On import the decomposition structure may not need to be preserved. The parts
may either merged into the <em>&quot;element&quot;</em>, or may become independent parts. The shape representation and property information however has to be preserved.</p>]]>
            </mvd:Body>
          </mvd:Definition>
        </mvd:Definitions>
      </mvd:ConceptTemplate>
    </mvd:SubTemplates>
  </mvd:Templates>
</mvd:mvdXML>
```

Page no.	Authors
15	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International


```

    </mvd:Body>
  </mvd:Definition>
</mvd:Definitions>
</mvd:ConceptTemplate>
<mvd:ConceptTemplate uuid="1b79f477-5787-4565-a60e-5fa28b756117" code="bSI-101" name="Composition" status="draft" version="1.0"
  owner="buildingSMART International" applicableSchema="IFC2x3">
  <mvd:Definitions>
    <mvd:Definition>
      <mvd:Body lang="en">
<![CDATA[<p>Provision of an aggregation structure where the <em>&quot;element&quot;;</em> is part of another element representing the aggregate. The <em>&quot;element&quot;;</em>
then provides the partial shape representation and it is positioned relative to the aggregate.</p>]]>
      </mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
</mvd:ConceptTemplate>
</mvd:SubTemplates>
</mvd:ConceptTemplate>
<mvd:ConceptTemplate uuid="32497bb6-8a8f-4e98-91ac-d512432764db" code="bSI-104" name="Classification" version="1.0" status="draft"
  owner="buildingSMART International" applicableSchema="IFC2x3">
  <mvd:Definitions>
    <mvd:Definition>
      <mvd:Body lang="en">
<![CDATA[<p>Provision of a classification reference as an link to an classification items and corresponding external classification system for the <em>&quot;element&quot;;</em>.
There can be more than one classification reference be provided for the <em>&quot;element&quot;;</em>.</p><p style="margin-left: 40px;">Note: Exchange requirements or local
conventions may determine the support of a particular classification system for a certain type of elements.</p><p style="margin-left: 40px;">Note: On import, the classification
reference may be shown as a property.</p>]]>
      </mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
</mvd:ConceptTemplate>
</mvd:Templates>
<mvd:Views>
  <mvd:ModelView uuid="f7c8d778-350f-434b-9195-f183bed35911" name="CoordinationView_V2.0" applicableSchema="IFC2x3" version="0.1">
    <mvd:Definitions>
      <mvd:Definition>
        <mvd:Body lang="en">
<![CDATA[The Coordination View Version 2.0 supports the design coordination between architectural, building service and structural engineering BIM applications.]]>
        </mvd:Body>
      </mvd:Definition>
    </mvd:Definitions>
    <mvd:ExchangeRequirements>
      <mvd:ExchangeRequirement name="Architecture" uuid="48b558c7-c5f4-41d1-9c69-76abecd85187">
        <mvd:Definitions>
          <mvd:Definition>
            <mvd:Body lang="en">
<![CDATA[The Exchange Requirement "Architecture" of the Coordination View Version 2.0 supports the design coordination between architectural application and architecture,
building service and structural engineering BIM applications. It identifies the particular export requirements from an architectural application.]]>
            </mvd:Body>
          </mvd:Definition>
        </mvd:Definitions>
      </mvd:ExchangeRequirement>
    </mvd:ExchangeRequirements>
  </mvd:ModelView>
</mvd:Views>

```

Page no.	Authors
16	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International


```

    </mvd:Definition>
  </mvd:Definitions>
</mvd:ExchangeRequirement>
<mvd:ExchangeRequirement name="Structural" uuid="61c75d12-11a4-41e5-859a-9890db51d700">
  <mvd:Definitions>
    <mvd:Definition>
      <mvd:Body lang="en">
<![CDATA[The Exchange Requirement "Structural" of the Coordination View Version 2.0 supports the design coordination between structural application and architecture, building
service and structural engineering BIM applications. It identifies the particular export requirements from an structural application.]]>
      </mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
</mvd:ExchangeRequirement>
<mvd:ExchangeRequirement name="BuildingService" uuid="0eff7a29-5e02-46fe-acca-8eb40f2dbf72">
  <mvd:Definitions>
    <mvd:Definition>
      <mvd:Body lang="en">
<![CDATA[The Exchange Requirement "BuildingService" of the Coordination View Version 2.0 supports the design coordination between building service application and architecture,
building service and structural engineering BIM applications. It identifies the particular export requirements from an building service application.]]>
      </mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
</mvd:ExchangeRequirement>
</mvd:ExchangeRequirements>
<mvd:Roots>
  <mvd:ConceptRoot uuid="30af73bf-3b38-4e4f-91a5-a987eb304cba" name="Beam" applicableRootEntity="IfcBeam">
    <mvd:Definitions>
      <mvd:Definition>
        <mvd:Body lang="en">
<![CDATA[The beam within the Coordination View Version 2.0 represents the architectural and structural design information of beams as a horizontal or nearly horizontal
longitudinal structure.]]>
        </mvd:Body>
      </mvd:Definition>
    </mvd:Definitions>
    <mvd:Concepts>
      <mvd:Concept uuid="55078f98-e68b-4f60-9cd1-4b6ffbe877f7" name="Aggregation" version="1.0" owner="buildingSMART International">
        <mvd:Template ref="02eefcc6-b26c-4f20-9c7c-c30ae74eb9a9"/>
        <mvd:Definitions>
          <mvd:Definition>
            <mvd:Body lang="en"><![CDATA[No further restriction of the concept template defined here.]]></mvd:Body>
          </mvd:Definition>
        </mvd:Definitions>
        <mvd:Requirements>
          <mvd:Requirement applicability="export" exchangeRequirement="48b558c7-c5f4-41d1-9c69-76abecd85187" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="61c75d12-11a4-41e5-859a-9890db51d700" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="0eff7a29-5e02-46fe-acca-8eb40f2dbf72" requirement="not relevant"/>
          <mvd:Requirement applicability="import" requirement="mandatory"/>
        </mvd:Requirements>
      </mvd:Concept>
    </mvd:Concepts>
  </mvd:ConceptRoot>
</mvd:Roots>

```

Page no.	Authors
17	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

    <mvd:SubConcepts>
      <mvd:Concept uuid="6bf8ed7b-5d78-4866-8e1d-b3a824f04354" name="Decomposition" version="1.0" owner="buildingSMART International">
        <mvd:Template ref="f6c8ea4c-e0aa-422c-a852-a1711a20d228"/>
        <mvd:Definitions>
          <mvd:Definition>
            <mvd:Body lang="en">
              <![CDATA[<p>Specific to <em>IfcBeam</em>: An <em>IfcBeam</em> with a composite profile definition is not handled as decomposition.</p><p style="margin-left: 40px;">An example for
              <em>IfcBeam</em> is an elemented <em>IfcBeam</em> that is decomposed into the <em>IfcBeam</em> section. <em>IfcMember</em>'s, <em>IfcFastener</em>'s, etc.. These decomposed
              elements have the geometric 'Body' shape representation, whereas the elemented <em>IfcBeam</em> does not have an own 'Body' shape representation.</p>]]>
            </mvd:Body>
          </mvd:Definition>
        </mvd:Definitions>
        <mvd:Requirements>
          <mvd:Requirement applicability="export" exchangeRequirement="48b558c7-c5f4-41d1-9c69-76abecd85187" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="61c75d12-11a4-41e5-859a-9890db51d700" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="0eff7a29-5e02-46fe-acca-8eb40f2dbf72" requirement="not relevant"/>
          <mvd:Requirement applicability="import" requirement="mandatory"/>
        </mvd:Requirements>
      </mvd:Concept>
      <mvd:Concept uuid="0369604b-79f1-40b9-b006-86057a7b4177" name="Composition" version="1.0" owner="buildingSMART International">
        <mvd:Template ref="1b79f477-5787-4565-a60e-5fa28b756117"/>
        <mvd:Definitions>
          <mvd:Definition>
            <mvd:Body lang="en">
              <![CDATA[<p>No further restriction of the concept template defined here.</p>
              <p style="margin-left: 40px;">An example for <em>IfcBeam</em> is an <em>IfcElementAssembly</em> such as a frame where the <em>IfcBeam</em> is a part of it. The <em>IfcBeam</em>
              is then positioned relative to this <em>IfcElementAssembly</em> and represents the shape of that part of the assembly.</p>]]>
            </mvd:Body>
          </mvd:Definition>
        </mvd:Definitions>
        <mvd:Requirements>
          <mvd:Requirement applicability="export" exchangeRequirement="48b558c7-c5f4-41d1-9c69-76abecd85187" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="61c75d12-11a4-41e5-859a-9890db51d700" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="0eff7a29-5e02-46fe-acca-8eb40f2dbf72" requirement="not relevant"/>
          <mvd:Requirement applicability="import" requirement="mandatory"/>
        </mvd:Requirements>
      </mvd:Concept>
    </mvd:SubConcepts>
  </mvd:Concept>
  <mvd:Concept name="Classification" uuid="4618fd33-e2d2-4f80-a776-cfe000d4d913" owner="buildingSMART International">
    <mvd:Template ref="32497bb6-8a8f-4e98-91ac-d512432764db"/>
    <mvd:Definitions>
      <mvd:Definition>
        <mvd:Body lang="en"><![CDATA[<p>No further restriction of the concept template defined here.</p>]]></mvd:Body>
      </mvd:Definition>
    </mvd:Definitions>
    <mvd:Requirements>
      <mvd:Requirement applicability="export" requirement="optional"/>
    </mvd:Requirements>
  </mvd:Concept>

```

Page no.	Authors
18	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

    <mvd:Requirement applicability="import" requirement="optional"/>
  </mvd:Requirements>
</mvd:Concept>
</mvd:Concepts>
</mvd:ConceptRoot>
<mvd:ConceptRoot uuid="a9a1bf86-bdf5-4e60-acea-a0ef282f1138" name="Column" applicableRootEntity="IfcColumn">
  <mvd:Definitions>
    <mvd:Definition>
      <mvd:Body lang="en">
<![CDATA[The column within the Coordination View Version 2.0 represents the architectural and structural design information of columns as a vertical or nearly vertical
longitudinal structure.]]>
      </mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
  <mvd:Concepts>
    <mvd:Concept name="Aggregation" uuid="39a535be-cab4-41a8-alda-33234143e4ab" version="1.0" owner="buildingSMART International">
      <mvd:Template ref="02eefcc6-b26c-4f20-9c7c-c30ae74eb9a9"/>
      <mvd:Definitions>
        <mvd:Definition>
          <mvd:Body lang="en"><![CDATA[<p>No further restriction of the concept template defined here.</p>]]></mvd:Body>
        </mvd:Definition>
      </mvd:Definitions>
      <mvd:Requirements>
        <mvd:Requirement applicability="export" exchangeRequirement="48b558c7-c5f4-41d1-9c69-76abecd85187" requirement="mandatory"/>
        <mvd:Requirement applicability="export" exchangeRequirement="61c75d12-11a4-41e5-859a-9890db51d700" requirement="mandatory"/>
        <mvd:Requirement applicability="export" exchangeRequirement="0eff7a29-5e02-46fe-acca-8eb40f2dbf72" requirement="not relevant"/>
        <mvd:Requirement applicability="import" requirement="mandatory"/>
      </mvd:Requirements>
    </mvd:SubConcepts>
    <mvd:Concept name="Decomposition" uuid="c3d31efb-4c99-4ff6-9038-0e78b6d96eb2" version="1.0" owner="buildingSMART International">
      <mvd:Template ref="f6c8ea4c-e0aa-422c-a852-a1711a20d228"/>
      <mvd:Definitions>
        <mvd:Definition>
          <mvd:Body lang="en"><![CDATA[<p>Specific to <em>IfcColumn</em>: An <em>IfcColumn</em> with a composite profile definition is not handled as
decomposition.</p><p style="margin-left: 40px;">An example for <em>IfcColumn</em> is an elemented <em>IfcColumn</em> that is decomposed into the <em>IfcColumn</em> section.
<em>IfcMember</em>'s, <em>IfcFastener</em>'s, etc. These decomposed elements have the geometric 'Body' shape representation, whereas the elemented <em>IfcColumn</em> does not
have an own 'Body' shape representation.</p>]]></mvd:Body>
          </mvd:Definition>
        </mvd:Definitions>
        <mvd:Requirements>
          <mvd:Requirement applicability="export" exchangeRequirement="48b558c7-c5f4-41d1-9c69-76abecd85187" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="61c75d12-11a4-41e5-859a-9890db51d700" requirement="mandatory"/>
          <mvd:Requirement applicability="export" exchangeRequirement="0eff7a29-5e02-46fe-acca-8eb40f2dbf72" requirement="not relevant"/>
          <mvd:Requirement applicability="import" requirement="mandatory"/>
        </mvd:Requirements>
      </mvd:Concept>
    <mvd:Concept name="Composition" uuid="feb4f32b-57dd-4b0e-83cf-23837b134838" version="1.0" owner="buildingSMART International">
      <mvd:Template ref="1b79f477-5787-4565-a60e-5fa28b756117"/>

```

Page no.	Authors
19	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

    <mvd:Definitions>
      <mvd:Definition>
        <mvd:Body lang="en">
          <![CDATA[<p>No further restriction of the concept template defined here.</p>
          <p style="margin-left: 40px;">An example for <em>IfcColumn</em> is an <em>IfcElementAssembly</em> such as a frame where the <em>IfcColumn</em> is a part of it. The
          <em>IfcColumn</em> is then positioned relative to this <em>IfcElementAssembly</em> and represents the shape of that part of the assembly.</p]]>
        </mvd:Body>
      </mvd:Definition>
    </mvd:Definitions>
    <mvd:Requirements>
      <mvd:Requirement applicability="export" exchangeRequirement="48b558c7-c5f4-41d1-9c69-76abecd85187" requirement="mandatory"/>
      <mvd:Requirement applicability="export" exchangeRequirement="61c75d12-11a4-41e5-859a-9890db51d700" requirement="mandatory"/>
      <mvd:Requirement applicability="export" exchangeRequirement="0eff7a29-5e02-46fe-acca-8eb40f2dbf72" requirement="not relevant"/>
      <mvd:Requirement applicability="import" requirement="mandatory"/>
    </mvd:Requirements>
  </mvd:Concept>
</mvd:SubConcepts>
</mvd:Concept>
<mvd:Concept name="Classification" uuid="4e204527-edac-4279-8397-35a934e448a8" version="1.0" owner="buildingSMART International">
  <mvd:Template ref="32497bb6-8a8f-4e98-91ac-d512432764db"/>
  <mvd:Definitions>
    <mvd:Definition>
      <mvd:Body lang="en"><![CDATA[<p>No further restriction of the concept template defined here.</p>]]></mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
  <mvd:Requirements>
    <mvd:Requirement applicability="export" exchangeRequirement="48b558c7-c5f4-41d1-9c69-76abecd85187" requirement="mandatory"/>
    <mvd:Requirement applicability="export" exchangeRequirement="61c75d12-11a4-41e5-859a-9890db51d700" requirement="mandatory"/>
    <mvd:Requirement applicability="export" exchangeRequirement="0eff7a29-5e02-46fe-acca-8eb40f2dbf72" requirement="not relevant"/>
    <mvd:Requirement applicability="import" requirement="optional"/>
  </mvd:Requirements>
</mvd:Concept>
</mvd:Concepts>
</mvd:ConceptRoot>
</mvd:Roots>
</mvd:ModelView>
</mvd:Views>
</mvd:mvdXML>

```

4.2 Concept Template with Constraints

The following is an example of a concept template for the ‘Axis’ representation used to define a path for elements such as beams, columns, walls, pipes, and ducts. This template is instantiated at concepts by enumerating available geometry types (e.g. *IfcBoundedCurve*, *IfcPolyline*), such that outer constraints (e.g. *RepresentationIdentifier*) are used to filter.

Page no.	Authors
20	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

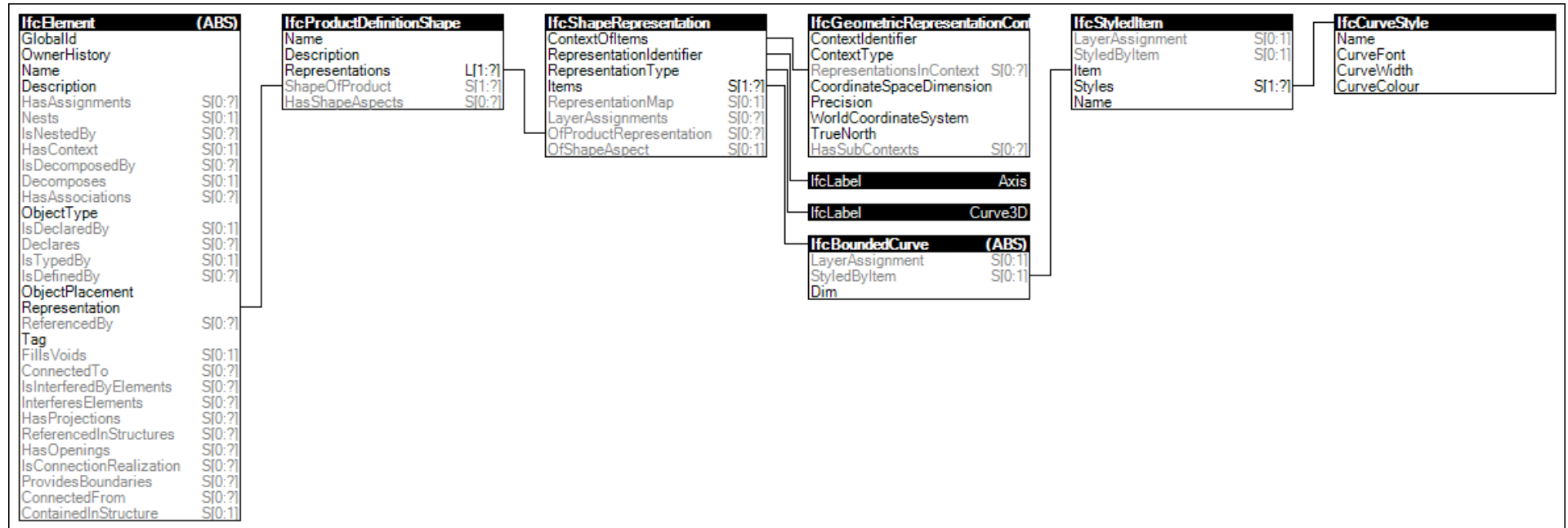


Figure 2: Example diagram created for a concept template for 'Axis representation'.

The following rules and constraints are set:

- The *AttributeRule* for *IfcElement.Representation* has 1:1 cardinality and an *EntityRule* for *IfcProductDefinitionShape* indicating that such value must be present; it would be invalid to have an instance of a different type such as *IfcMaterialDefinitionRepresentation*. If the requirement for a particular exchange is 'mandatory', then the value must not be NULL. If the requirement for a particular exchange is 'optional', then the value may be NULL or match the rules the same as if it were mandatory. If the requirement for a particular exchange is 'excluded', then the value must be NULL. Attributes without rules defined observe the same requirements as the schema (it does not mean a NULL value).
- The *AttributeRule* for *IfcProductDefinitionShape.Representations* has 1:1 cardinality and an *EntityRule* for *IfcShapeRepresentation* indicating that the LIST must contain one instance matching the configuration. An empty list would be invalid (both due to the schema-defined cardinality as well as the rule).
- The *AttributeRule* for *IfcShapeRepresentation.RepresentationIdentifier* has 1:1 cardinality and an *EntityRule* for *IfcLabel* with a Constraint indicating the value must equal 'Axis'. Since it is an outer constraint (the *RuleID* of referencing concepts maps to *IfcShapeRepresentation.Items*), a list containing a shape

representation with another value (such as 'Body') would not violate the rule (such instance would not be subject to the rule), however a list that does not contain one 'Axis' representation would be in violation, as would a list containing multiple 'Axis' representations.

- The *AttributeRule* for *IfcShapeRepresentation.Items* has 1:1 cardinality and an *EntityRule* for *IfcBoundedCurve* indicating that the SET must contain a single *IfcBoundedCurve* subtype such as *IfcPolyline*. A set containing multiple items would be invalid.
- The *AttributeRule* for *IfcBoundedCurve.StyledByItem* has 0:1 cardinality and an *EntityRule* for *IfcStyledItem* indicating that if present, there must be a single *IfcStyledItem* assigned.
- The *AttributeRule* for *IfcStyledItem.Styles* has 1:1 cardinality and an *EntityRule* for *IfcCurveStyle* indicating that there must be a single *IfcCurveStyle* assigned (that is, if there is an *IfcStyledItem*).

4.3 Concept Template with Parameters

The following is an example of a concept template for ports.

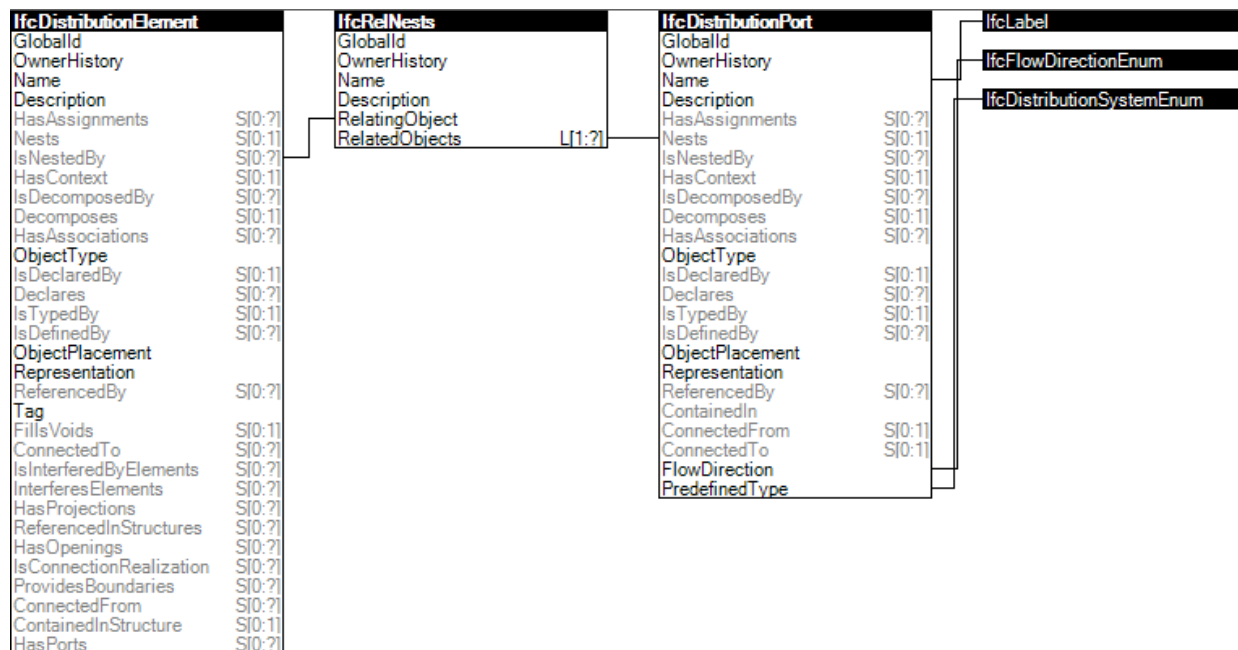


Figure 3: Example diagram created for a concept template for 'Axis representation'.

The mvdXML definition of that *mvd:ConceptTemplate* with rules is:

```
<mvd:ConceptTemplate uuid="bafc93b7-d0e2-42d8-84cf-5da20ee1480a" code="" name="Port" applicableSchema="IFC4"
applicableEntity="IfcDistributionElement">
  <mvd:Definitions>
    <mvd:Definition >
      <mvd:Body>
<![CDATA[<p>Distribution ports are defined by <i>IfcDistributionPort</i> and attached by the <i>IfcRelNests</i> relationship. Ports are specific to
the PredefinedType as follows indicated by the <i>IfcDistributionPort</i> <i>Name</i>, <i>PredefinedType</i>, and <i>FlowDirection</i>:</p>]]>
      </mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
  <mvd:Rules>
    <mvd:AttributeRule AttributeName="PredefinedType" RuleID="PredefinedType"/>
    <mvd:AttributeRule AttributeName="IsNestedBy" >
      <mvd:EntityRules>
        <mvd:EntityRule EntityName="IfcRelNests" >
          <mvd:AttributeRules>
            <mvd:AttributeRule AttributeName="RelatedObjects" >
              <mvd:EntityRules>
                <mvd:EntityRule EntityName="IfcDistributionPort" RuleID="Port">
                  <mvd:AttributeRules>
                    <mvd:AttributeRule AttributeName="Name" RuleID="Name"/>
                    <mvd:AttributeRule AttributeName="PredefinedType" RuleID="Type"/>
                    <mvd:AttributeRule AttributeName="FlowDirection" RuleID="Flow"/>
                  </mvd:AttributeRules>
                </mvd:EntityRule>
              </mvd:EntityRules>
            </mvd:AttributeRule>
          </mvd:AttributeRules>
        </mvd:EntityRule>
      </mvd:EntityRules>
    </mvd:AttributeRule>
  </mvd:Rules>
</mvd:ConceptTemplate>
```

4.4 Concept Template with Referenced Values

The following is an example of a template using referenced values and constraints.

```
<mvd:ConceptTemplate uuid="abcd1234-d0e2-42d8-84cf-5da20ee1480a" code="" name="Beam" applicableSchema="IFC4" applicableEntity="IfcBeam">
  <mvd:Rules>
    <mvd:AttributeRule AttributeName="HasAssociations" >
      <mvd:EntityRules>
        <mvd:EntityRule EntityName="IfcRelAssociatesMaterial" >
```

Page no.	Authors
23	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

<mvd:AttributeRules>
  <mvd:AttributeRule AttributeName="RelatedMaterial" >
    <mvd:EntityRules>
      <mvd:EntityRule EntityName="IfcMaterialProfileSetUsage">
        <mvd:AttributeRules>
          <mvd:AttributeRule AttributeName="MaterialProfile">
            <mvd:ValueConstraints>
              <mvd:Constraint Expression="#MP">
            </mvd:ValueConstraints>
          </mvd:AttributeRule>
        </mvd:AttributeRules>
      </mvd:EntityRule>
    </mvd:EntityRules>
  </mvd:AttributeRule>
</mvd:AttributeRules>
</mvd:EntityRule>
</mvd:EntityRules>
</mvd:AttributeRule>
</mvd:AttributeRules>
</mvd:EntityRule>
</mvd:EntityRules>
</mvd:AttributeRule>
<mvd:AttributeRule AttributeName="Representation" >
  <mvd:EntityRules>
    <mvd:EntityRule EntityName="IfcProductDefinitionShape" >
      <mvd:AttributeRules>
        <mvd:AttributeRule AttributeName="Representations" >
          <mvd:EntityRules>
            <mvd:EntityRule EntityName="IfcShapeRepresentation">
              <mvd:AttributeRules>
                <mvd:AttributeRule AttributeName="RepresentationIdentifier">
                  <mvd:ValueConstraints>
                    <mvd:Constraint Expression="Body">
                  </mvd:ValueConstraints>
                </mvd:AttributeRule>
                <mvd:AttributeRule AttributeName="RepresentationType">
                  <mvd:ValueConstraints>
                    <mvd:Constraint Expression="SweptSolid">
                  </mvd:ValueConstraints>
                </mvd:AttributeRule>
                <mvd:AttributeRule AttributeName="Items">
                  <mvd:EntityRules>
                    <mvd:EntityRule EntityName="IfcExtrudedAreaSolid">
                      <mvd:AttributeRules>
                        <mvd:AttributeRule AttributeName="SweptProfile">
                          <mvd:ValueConstraints>
                            <mvd:Constraint Expression="#MP">
                          </mvd:ValueConstraints>
                        </mvd:AttributeRule>
                      </mvd:AttributeRules>
                    </mvd:EntityRule>
                  </mvd:EntityRules>
                </mvd:AttributeRule>
              </mvd:EntityRules>
            </mvd:EntityRule>
          </mvd:EntityRules>
        </mvd:AttributeRule>
      </mvd:AttributeRules>
    </mvd:EntityRule>
  </mvd:EntityRules>
</mvd:AttributeRule>

```

Page no.	Authors
24	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International


```

        </mvd:AttributeRule>
      </mvd:AttributeRules>
    </mvd:EntityRule>
  </mvd:EntityRules>
</mvd:AttributeRule>
</mvd:AttributeRules>
</mvd:EntityRule>
</mvd:EntityRules>
</mvd:AttributeRule>
</mvd:Rules>
</mvd:ConceptTemplate>

```

4.5 Concept Root using Template Values

The following is an example of a concept root, with concept and template rules. The *TemplateRule* has a *RuleID* of “Port” corresponding to the *EntityRule* for *IfcDistributionPort* which is to be instantiated. The parameters (Name, Type, Source) refer to *AttributeRule*’s with the corresponding values substituted.

```

<mvd:ConceptRoot uuid="8b949664-a5df-4bfc-922c-4a486c41d756" name="Sensor" applicableRootEntity="IfcSensor">
  <mvd:Definitions>
    <mvd:Definition>
      <mvd:Body>
        <![CDATA[<p>A sensor is a device that measures a physical quantity and converts it into a signal which can be read by an observer or by an
instrument.</p>]]>
      </mvd:Body>
    </mvd:Definition>
  </mvd:Definitions>
  <mvd:Concepts>
    <mvd:Concept uuid="a4fa348c-a025-4a02-abfd-c42fd0901540" name="Port">
      <mvd:Template ref="bafc93b7-d0e2-42d8-84cf-5da20ee1480a"/>
      <mvd:Rules>
        <mvd:TemplateRule RuleID="Port" Parameters="Name=Output;Type=SIGNAL;Flow=SOURCE;" Description="Transmits signal."/>
      </mvd:Rules>
    </mvd:Concept>
  </mvd:Concepts>
</mvd:ConceptRoot>

```

Page no.	Authors
25	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

5 XSD Listing

The following listing the computer-interpretable XSD schema for mvdXML.

```
<?xml version="1.0" encoding="utf-8"?>
<!-- Mit XMLSpy v2012 sp1 (x64) (http://www.altova.com) von Thomas Liebich (AEC3 Deutschland GmbH) bearbeitet -->
<xs:schema xmlns:mvd="http://buildingsmart-tech.org/mvdXML/mvdXML1-0" xmlns:xs="http://www.w3.org/2001/XMLSchema"
targetNamespace="http://buildingsmart-tech.org/mvdXML/mvdXML1-0" elementFormDefault="qualified">
  <xs:element name="mvdXML" type="mvd:mvdXML">
    <xs:unique name="unique-uuids">
      <xs:selector xpath="./mvd:ModelView | ./mvd:ConceptTemplate | ./mvd:ConceptRoot | ./mvd:ExchangeRequirement | ./mvd:Concept"/>
      <xs:field xpath="@uuid"/>
    </xs:unique>
    <xs:key name="TemplateKey">
      <xs:selector xpath="./mvd:ConceptTemplate"/>
      <xs:field xpath="@uuid"/>
    </xs:key>
    <xs:keyref name="ConceptToTemplateRef" refer="mvd:TemplateKey">
      <xs:selector xpath="./mvd:Template"/>
      <xs:field xpath="@ref"/>
    </xs:keyref>
    <xs:key name="ExchangeRequirementKey">
      <xs:selector xpath="./mvd:ExchangeRequirement"/>
      <xs:field xpath="@uuid"/>
    </xs:key>
    <xs:keyref name="RequirementToExchangeRequirementRef" refer="mvd:ExchangeRequirementKey">
      <xs:selector xpath="./mvd:Requirement"/>
      <xs:field xpath="@exchangeRequirement"/>
    </xs:keyref>
  </xs:element>
  <xs:complexType name="mvdXML">
    <xs:sequence>
      <xs:element name="Templates" minOccurs="1">
        <xs:complexType>
          <xs:sequence>
            <xs:element ref="mvd:ConceptTemplate" maxOccurs="unbounded"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
      <xs:element name="Views" minOccurs="0">
        <xs:complexType>
          <xs:sequence>
            <xs:element ref="mvd:ModelView" maxOccurs="unbounded">
            </xs:element>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

Page no.	Authors
26	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

    </xs:complexType>
  </xs:element>
</xs:sequence>
<xs:attributeGroup ref="mvd:identity"/>
</xs:complexType>
<xs:element name="ModelView" type="mvd:ModelView"/>
<xs:complexType name="ModelView">
  <xs:sequence>
    <xs:element name="BaseView" minOccurs="0">
      <xs:complexType>
        <xs:attribute name="ref" type="mvd:uuid"/>
        <xs:attribute name="href" type="xs:anyURI"/>
      </xs:complexType>
    </xs:element>
    <xs:element name="Definitions" type="mvd:Definitions" minOccurs="0"/>
    <xs:element name="ExchangeRequirements" minOccurs="0">
      <xs:complexType>
        <xs:sequence>
          <xs:element ref="mvd:ExchangeRequirement" maxOccurs="unbounded"/>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
    <xs:element name="Roots" minOccurs="1">
      <xs:complexType>
        <xs:sequence>
          <xs:element ref="mvd:ConceptRoot" maxOccurs="unbounded"/>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
  <xs:attributeGroup ref="mvd:identity"/>
  <xs:attribute name="applicableSchema" type="xs:normalizedString"/>
</xs:complexType>
<xs:element name="ConceptTemplate" type="mvd:ConceptTemplate"/>
<xs:complexType name="ConceptTemplate">
  <xs:sequence>
    <xs:element name="Definitions" type="mvd:Definitions" minOccurs="0"/>
    <xs:element name="Rules" type="mvd:Rules" minOccurs="0"/>
    <xs:element name="SubTemplates" minOccurs="0">
      <xs:complexType>
        <xs:sequence>
          <xs:element ref="mvd:ConceptTemplate" maxOccurs="unbounded"/>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
</xs:complexType>

```

Page no.	Authors
27	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

<xs:attributeGroup ref="mvd:identity"/>
<xs:attribute name="applicableSchema" type="xs:normalizedString" use="required"/>
<xs:attribute name="applicableEntity" use="optional">
  <xs:simpleType>
    <xs:list itemType="xs:normalizedString"/>
  </xs:simpleType>
</xs:attribute>
</xs:complexType>
<xs:element name="ConceptRoot" type="mvd:ConceptRoot"/>
<xs:complexType name="ConceptRoot">
  <xs:sequence>
    <xs:element name="Definitions" type="mvd:Definitions" minOccurs="0"/>
    <xs:element name="Requirements" type="mvd:Requirements" minOccurs="0"/>
    <xs:element name="Concepts" minOccurs="0">
      <xs:complexType>
        <xs:sequence>
          <xs:element ref="mvd:Concept" maxOccurs="unbounded"/>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:attributeGroup ref="mvd:identity"/>
<xs:attribute name="applicableRootEntity" type="xs:normalizedString" use="optional"/>
</xs:complexType>
<xs:element name="Concept" type="mvd:Concept"/>
<xs:complexType name="Concept">
  <xs:sequence>
    <xs:element name="Template">
      <xs:complexType>
        <xs:attribute name="ref" type="mvd:uuid"/>
        <xs:attribute name="href" type="xs:anyURI"/>
      </xs:complexType>
    </xs:element>
    <xs:element name="Definitions" type="mvd:Definitions" minOccurs="0"/>
    <xs:element name="Requirements" type="mvd:Requirements" minOccurs="0"/>
    <xs:element name="Rules" type="mvd:Rules" minOccurs="0"/>
    <xs:element name="SubConcepts" minOccurs="0">
      <xs:complexType>
        <xs:sequence>
          <xs:element ref="mvd:Concept" maxOccurs="unbounded"/>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:attributeGroup ref="mvd:identity"/>
</xs:complexType>
<xs:element name="ExchangeRequirement" type="mvd:ExchangeRequirement"/>

```

Page no.	Authors
28	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

<xs:complexType name="ExchangeRequirement">
  <xs:sequence>
    <xs:element name="Definitions" type="mvd:Definitions" minOccurs="0"/>
  </xs:sequence>
  <xs:attributeGroup ref="mvd:identity"/>
  <xs:attribute ref="applicability" use="optional"/>
</xs:complexType>
<xs:complexType name="Definitions">
  <xs:sequence>
    <xs:element name="Definition" minOccurs="1" maxOccurs="unbounded">
      <xs:complexType>
        <xs:sequence>
          <xs:element name="Body" maxOccurs="1">
            <xs:complexType>
              <xs:simpleContent>
                <xs:extension base="xs:string">
                  <xs:attribute name="lang" type="xs:language" use="optional" default="en"/>
                </xs:extension>
              </xs:simpleContent>
            </xs:complexType>
          </xs:element>
          <xs:element name="Link" nillable="true" minOccurs="0" maxOccurs="unbounded">
            <xs:complexType>
              <xs:attribute name="lang" type="xs:language" use="optional" default="en"/>
              <xs:attribute name="title" type="xs:normalizedString" use="optional"/>
              <xs:attribute name="category" use="optional">
                <xs:simpleType>
                  <xs:restriction base="xs:string">
                    <xs:enumeration value="definition"/>
                    <xs:enumeration value="agreement"/>
                    <xs:enumeration value="diagram"/>
                    <xs:enumeration value="instantiation"/>
                    <xs:enumeration value="example"/>
                  </xs:restriction>
                </xs:simpleType>
              </xs:attribute>
              <xs:attribute name="href" type="xs:anyURI" use="required"/>
            </xs:complexType>
          </xs:element>
        </xs:sequence>
      </xs:complexType>
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:complexType name="Requirements">
  <xs:sequence>
    <xs:element name="Requirement" maxOccurs="unbounded">
      <xs:complexType>

```

Page no.	Authors
29	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

    <xs:simpleContent>
      <xs:extension base="xs:string">
        <xs:attribute name="exchangeRequirement" type="mvd:uuid" use="optional"/>
        <xs:attribute ref="applicability" use="required"/>
        <xs:attribute name="requirement" use="optional">
          <xs:simpleType>
            <xs:restriction base="xs:string">
              <xs:enumeration value="mandatory"/>
              <xs:enumeration value="optional"/>
              <xs:enumeration value="not relevant"/>
              <xs:enumeration value="excluded"/>
            </xs:restriction>
          </xs:simpleType>
        </xs:attribute>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
<xs:simpleType name="uuid">
  <xs:restriction base="xs:string">
    <xs:length value="36" fixed="true"/>
    <xs:pattern value="[a-f0-9]{8}-[a-f0-9]{4}-[a-f0-9]{4}-[a-f0-9]{4}-[a-f0-9]{12}"/>
  </xs:restriction>
</xs:simpleType>
<xs:attributeGroup name="identity">
  <xs:attribute name="uuid" type="mvd:uuid" use="required"/>
  <xs:attribute name="name" type="xs:normalizedString" use="required"/>
  <xs:attribute name="code" type="xs:normalizedString" use="optional"/>
  <xs:attribute name="version" type="xs:normalizedString" use="optional"/>
  <xs:attribute name="status" type="xs:normalizedString" use="optional"/>
  <xs:attribute name="author" type="xs:normalizedString" use="optional"/>
  <xs:attribute name="owner" type="xs:normalizedString" use="optional"/>
  <xs:attribute name="copyright" type="xs:anyURI" use="optional"/>
</xs:attributeGroup>
<xs:attribute name="applicability">
  <xs:simpleType>
    <xs:restriction base="xs:string">
      <xs:enumeration value="export"/>
      <xs:enumeration value="import"/>
    </xs:restriction>
  </xs:simpleType>
</xs:attribute>
<xs:element name="AbstractRule" type="mvd:AbstractRule" abstract="true"/>
<xs:complexType name="AbstractRule" abstract="true">
  <xs:attribute name="RuleID" type="xs:normalizedString" use="optional"/>
  <xs:attribute name="Description" type="xs:string"/>

```

Page no.	Authors
30	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

</xs:complexType>
<xs:complexType name="Rules">
  <xs:sequence>
    <xs:element ref="mvd:AbstractRule" maxOccurs="unbounded">
    </xs:element>
  </xs:sequence>
</xs:complexType>
<xs:element name="AttributeRule" type="mvd:AttributeRule" substitutionGroup="mvd:AbstractRule"/>
<xs:complexType name="AttributeRule">
  <xs:complexContent>
    <xs:extension base="mvd:AbstractRule">
      <xs:sequence>
        <xs:element name="EntityRules" minOccurs="0">
          <xs:complexType>
            <xs:sequence maxOccurs="unbounded">
              <xs:element ref="mvd:EntityRule"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="Constraints" minOccurs="0">
          <xs:complexType>
            <xs:sequence maxOccurs="unbounded">
              <xs:element name="Constraint">
                <xs:complexType>
                  <xs:attribute name="Expression" type="xs:string" use="required"/>
                </xs:complexType>
              </xs:element>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
      <xs:attribute name="AttributeName" type="xs:string" use="required"/>
      <xs:attribute name="Cardinality" use="optional">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:enumeration value="_asSchema"/>
            <xs:enumeration value="Zero"/>
            <xs:enumeration value="ZeroToOne"/>
            <xs:enumeration value="One"/>
            <xs:enumeration value="OneToMany"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:element name="EntityRule" substitutionGroup="mvd:AbstractRule">
  <xs:complexType>

```

Page no.	Authors
31	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

```

    <xs:complexContent>
      <xs:extension base="mvd:EntityRule"/>
    </xs:complexContent>
  </xs:complexType>
</xs:element>
<xs:complexType name="EntityRule">
  <xs:complexContent>
    <xs:extension base="mvd:AbstractRule">
      <xs:sequence>
        <xs:element name="AttributeRules" minOccurs="0">
          <xs:complexType>
            <xs:sequence maxOccurs="unbounded">
              <xs:element ref="mvd:AttributeRule"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="EntityRules" minOccurs="0">
          <xs:complexType>
            <xs:sequence maxOccurs="unbounded">
              <xs:element ref="mvd:EntityRule"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
        <xs:element name="Constraints" minOccurs="0">
          <xs:complexType>
            <xs:sequence maxOccurs="unbounded">
              <xs:element name="Constraint">
                <xs:complexType>
                  <xs:attribute name="Expression" type="xs:string" use="required"/>
                </xs:complexType>
              </xs:element>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
      <xs:attribute name="EntityName" type="xs:string" use="required"/>
      <xs:attribute name="Cardinality" use="optional">
        <xs:simpleType>
          <xs:restriction base="xs:string">
            <xs:enumeration value="_asSchema"/>
            <xs:enumeration value="Zero"/>
            <xs:enumeration value="ZeroToOne"/>
            <xs:enumeration value="One"/>
            <xs:enumeration value="OneToMany"/>
          </xs:restriction>
        </xs:simpleType>
      </xs:attribute>
    </xs:extension>
  </xs:complexType>

```

Page no.	Authors
32	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International


```

    </xs:complexContent>
  </xs:complexType>
  <xs:element name="TemplateRule" substitutionGroup="mvd:AbstractRule">
    <xs:complexType>
      <xs:complexContent>
        <xs:extension base="mvd:TemplateRule"/>
      </xs:complexContent>
    </xs:complexType>
  </xs:element>
  <xs:complexType name="TemplateRule">
    <xs:complexContent>
      <xs:extension base="mvd:AbstractRule">
        <xs:attribute name="Parameters" type="xs:string" use="required"/>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

Page no.	Authors
33	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International

6 Acknowledgment

Parts of this work have been supported by the research project Mefisto (<http://www.mefisto-bau.de/>), partially funded by the German Ministry of Education and Research, and by the research project HESMOS (<http://www.hesmos.eu/>), partially funded by the European Union. The co-authors Matthias Weise and Thomas Liebich (AEC3) would not have been able to significantly contribute to this open international standardization work of mvdXML without this generous support of the aforementioned research projects.

Page no.	Authors
34	Chipman, T; Liebich, T; Weise, M - Model Support Group of buildingSMART International